

Connor Savugot

B.S. in Computer Engineering | North Carolina State University | Graduated May 2026

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August 14, 2026

Re: Embedded Firmware Engineer

Dear Hiring Manager,

I am applying for the Embedded Firmware Engineer position at your organization. I graduated from North Carolina State University in May 2026 with a B.S. in Computer Engineering and joined AEGIS Power Systems that month as an Embedded Firmware Engineer after completing two summer internships there. My focus is low-level firmware and hardware integration: writing embedded C/C++ for microcontrollers and DSPs, bringing up peripherals, and carrying features from datasheets and schematics through bench validation and documented handoff.

At AEGIS, I develop and bring up 16-bit dsPIC33CK firmware and board peripherals, isolating register, timing, hardware, and configuration faults with focused tests and bench instruments. I implemented and validated Tier 2 management firmware for a VPX power supply, following VITA 46.11/IPMI 2.0 over I2C-based IPMB for chassis sensor, inventory, and event reporting. I also integrate battery hardware and CAN-connected devices with application software. Where products include Embedded Linux, I support TI AM62 systems built with Yocto/Arago, including kernels, device trees, boot diagnostics, systemd services, Docker, and cross-platform tooling.

Across my AEGIS internships, I built firmware for a PIC32 running FreeRTOS for communication and update coordination and a TMS320 DSP with bare-metal C firmware using TI peripheral libraries for programmable bidirectional power supplies. For one PIC32/TMS320 product, I built a verified firmware-update and recovery path spanning host-side C++/Python tools, external flash storage, processor-to-processor programming, and boot control. I also integrate and debug SPI, UART/RS-232, I2C, and CAN systems, including Classical CAN (CAN 2.0), CAN FD, J1939, and DroneCAN.

For my NC State senior design project, I helped architect and validate a custom-PCB battery-management and CAN telemetry system. I translated component tradeoffs into subsystem interfaces and tests while coordinating design reviews, risk tracking, documentation, and bring-up across the team.

Teaching has strengthened those same engineering skills. Across three semesters supporting ECE 306 and two semesters leading IEEE Open Project Space 2, I delivered supplemental lectures, led laboratories, debugged firmware and hardware, and coached students through open-ended projects. I also coordinated and taught through-hole and surface-mount soldering workshops. These roles made me a clearer technical communicator and a more disciplined debugger.

I would bring low-level embedded C/C++, real-time control, and hardware-integration experience to your organization. I am especially interested in teams building dependable embedded products where hardware, firmware, and system software must be validated together. I would welcome the opportunity to discuss how I could contribute to your team.

Sincerely,

Connor Savugot